

REQUEST FOR PROPOSAL - DEMO PACKAGE

LNG Compressor Recycle Valve Package

Train 4 Reliability and Anti-Surge Modernization

FICTIONAL CUSTOMER PROFILE

Aurora Gulf LNG Export Terminal

INDUSTRY

LNG / Gas Processing

ISSUE DATE

29 May 2026

BID DUE DATE

19 June 2026

COMMERCIAL CONTEXT

Estimated package value: INR 28.5 Cr

PACKAGE TYPE

Severe-service control valve proposal

Demo-safe fictional RFP package created for WinsProposal. Customer names are fictional industry-style accounts and do not represent official documents, endorsements, trademarks, or partnerships.

1. Purchaser Profile and Project Background

Aurora Gulf LNG Export Terminal is a fictional global LNG operator profile used to represent a mature, high-reliability process owner with rigorous engineering, documentation, and inspection expectations.

The project covers replacement and upgrade of compressor recycle / anti-surge control valves serving LNG refrigerant and feed-gas compression systems. The package is safety and availability critical. The purchaser requires a technically compliant proposal that demonstrates severe-service application understanding, fast response capability, acoustic risk awareness, actuator/accessory readiness, and clear inspection deliverables.

Proposal expectation: Supplier response shall show engineering proposal intelligence: extracted requirements, compliance/TBE readiness, severe-service risk recognition, knowledge reuse, deviations, review owners, and management-ready documentation.

2. Scope of Supply

- Design, manufacture, test, document, preserve, and deliver severe-service compressor recycle control valves.
- Provide actuator and control accessory packages suitable for anti-surge service.
- Provide preliminary valve sizing basis, severe-service risk narrative, acoustic/noise assumptions, and engineering validation checklist.
- Provide compliance matrix and technical bid evaluation responses by line item.
- Provide QA/QC documentation, test certificates, inspection records, and final manufacturing data book.

3. Line Items

Tag / Ref.	Item	Qty	Indicative Size / Class	Service Notes
XV-CRV-9101A/B/C	Compressor recycle control valve	3	NPS 10 x 12, ASME 900	Fast-acting gas recycle, high pressure drop
ASP-9101A/B/C	Anti-surge accessory package	3	Pneumatic accessories	Positioner, solenoids, boosters, limit switches, airset
DOC-9101	Engineering and QA dossier	1 lot	MDR package	Datasheets, calculations summary, ITP, QAP, certificates

4. Process Conditions

Input	Requirement / Value
Fluid	Natural gas rich LNG process gas
Inlet pressure	92 barg normal, 108 barg design
Outlet pressure	34 barg normal
Temperature	-28 C to 45 C
Flow cases	Normal recycle, startup bypass, compressor trip
Required response	Fast stroke for anti-surge duty; supplier to state assumptions
Leakage class	Class IV minimum; tighter class subject to final engineering review

5. Technical Specification

- Valve shall be suitable for severe-service gas pressure letdown with potential choked-flow and high aerodynamic noise.
- Supplier shall propose multi-stage or multi-path severe-service trim where required by final validated sizing.
- Supplier shall identify outlet velocity, acoustic fatigue, trim exit velocity, and vibration risk assumptions.
- Actuator shall be sized for fail action, trip condition, shutoff pressure, instrument air pressure, and response requirements.
- Accessory package shall include smart positioner, solenoid arrangement, airset, limit switches, tubing, and boosters or quick exhaust devices where validated.
- Supplier shall clearly separate proposal-stage indicative engineering from final certified sizing/design.

6. Compliance Requirements

- ISA 75.01 / IEC 60534 awareness for control valve sizing.
- ASME B16.34 for pressure-temperature considerations.
- Project anti-surge control philosophy and compressor OEM interface requirements.
- Project noise specification and acoustic fatigue review requirements.
- Client inspection and manufacturing data record requirements.

7. Inspection and Testing Requirements

- Hydrostatic shell test.
- Seat leakage test.
- Functional stroke test with actuator/accessory package.
- Accessory loop check and fail-action demonstration.
- PMI for pressure-retaining components.
- Client witness points for pressure test, functional test, and final inspection.

8. Required Proposal-Stage Engineering Intelligence

Engineering Narrative Required

- Extract process conditions and classify application/service.
- Provide preliminary valve configuration narrative.
- Provide severe-service risk assessment.
- Provide cavitation/flashing/choked-flow narrative where relevant.
- Provide noise/vibration risk narrative.
- Provide trim, material, actuator, and accessory notes.

Calculation Summary Required

- State process inputs and assumptions.
- Classify pressure drop severity.
- Provide indicative Cv/Kv placeholder only where safe.
- List risk flags and missing data.
- List validation checklist and standards awareness.
- State that final sizing/design requires engineer validation.

9. Technical Bid Evaluation Template

Evaluation Tag	Purchaser Requirement
Material	Pressure boundary material shall match LNG low-temperature line class and ASME pressure-temperature basis.
Trim	Severe-service multi-stage low-noise trim expected; final selection by validated sizing.
Actuator	Fast-response pneumatic actuator; supplier to define response assumptions.
Accessories	Smart positioner, solenoids, limit switches, booster/quick exhaust if required.
Testing	Hydrotest, seat leakage, functional stroke, accessory loop check, witness inspection.
Documentation	Datasheet, GA, ITP, QAP, MTCs, test certificates, deviation register, O&M manual.

10. Evaluation Criteria

Criterion	Weight
Technical compliance	40%
Severe-service application understanding	20%
Delivery and inspection readiness	15%
Documentation quality	10%
Commercial value	15%

11. Possible Deviations and Clarifications

- Final acoustic guarantee requires confirmed flow cases, gas composition, pipe geometry, and project noise limit.
- Final Cv/Kv, trim model, and stage count require company-approved sizing tools.
- Final actuator response claim requires confirmed instrument air pressure and anti-surge control philosophy.

12. Engineering Review Points

- Confirm compressor anti-surge response requirement and trip cases.
- Confirm gas composition, molecular weight, compressibility, and maximum/minimum temperatures.
- Confirm outlet pipe geometry, allowable sound pressure level, and acoustic fatigue criteria.

- Confirm fail action, instrument air pressure, and solenoid logic.
- Confirm whether final proposal requires guaranteed noise or response time.

13. Commercial and Delivery Assumptions

- Supplier shall state drawing submission lead time, manufacturing lead time, test duration, and shipping readiness.
- Supplier shall identify long-lead materials, special trims, actuator accessories, and inspection hold points that may affect schedule.
- Supplier shall provide itemized pricing by line item and list optional compliance or testing costs separately.
- Supplier shall include a deviation register with technical, commercial, and schedule impact.
- Proposal validity shall be not less than 60 days from bid due date.

Mandatory wording: Any engineering analysis in the supplier proposal shall clearly state: "Preliminary proposal-stage engineering estimate. Final sizing/design must be validated by qualified engineers using company-approved tools and standards."